17.7.4. Mass separator procedures

In the ISOLDE (Isotope Separation On Line DEvelopment) collaboration at CERN, the target material is fixed in a solid but porous chemical matrix and irradiated by protons or heavy ions. The products react with the target matrix, and those with the "right" chemistry diffuse from the target (which may be heated to increase the diffusion rate) and are swept by a gas stream into the ionization chamber of a mass spectrometer where the products are separated. Figure 17.9 shows the schematic for such a system: At 1, a 600 MeV proton beam is focused on the target 2. The beam of spallation products is accelerated to 60 keV, focused by magnet 3 and separated by electrostatic deflectors at 4 to yield a number of beams which are analyzed by various techniques (nuclear spectroscopy, laser spectroscopy, atomic beam magnetic resonance, range measurements, etc).

17.8. EXERCISES

17.1. ^{24}Na is produced through the reaction ^{26}Mg(d,α)^{24}Na. A 0.2 mm thick magnesium foil is irradiated for 1 h by a current of 130 μA of 22 MeV D^+ ions in a cyclotron. The foil has a much larger area than the cross-section of the beam. What is the specific activity of ^{24}Na if the magnesium foil (3 cm^2) contains 0.003% Na and σ for the reaction is assumed to be 25 mb?

17.2. Oxygen can be determined through the reaction ^{16}O(n,p)^{16}N $\rightarrow$ (β,γ 7 s); σ_{np} at 14 MeV n is 49 mb. 3.982 g of a fatty acid were irradiated for 20 s in 4×10^{12} n m^{-2} s^{-1}. After the irradiation the sample was rapidly transferred with a rabbit system to a scintillation detector which had an efficiency of 1.1% for the ^{16}N γ-rays ($\sim$6 MeV). Exactly 8 s after the end of the irradiation, the sample was counted for 1 min, yielding 13 418 counts above background. What was the oxygen fraction of the sample?

17.3. ^{199}Au can be formed through two successive n,γ-reactions on ^{197}Au (100% in nature). If 1 g ^{197}Au is irradiated with 10^{18} n m^{-2} s^{-1} during 30 h, what will the disintegration rate of ^{199}Au be at the end of the irradiation? The chain of events to be considered is:

$$^{197}\text{Au}\ (\sigma_{n,\gamma}\ 98.8\ \text{b}) \rightarrow {}^{198}\text{Au}\left(t_{1/2}\ 2.694\ \text{d},\ \sigma_{n,\gamma}\ 25\,100\ \text{b}\right) \rightarrow$$
$$^{199}\text{Au}\left(t_{1/2}\ 3.139\ \text{d},\ \sigma_{n,\gamma}\ 30\ \text{b}\right) \rightarrow \text{Neglect self shielding.}$$

17.4. ^{246}Cm has a half-life of 4 730 y. It can be obtained through neutron capture in ^{245}Cm, which has a half-life of 8 500 y; the reaction cross-section is 345 b. Both isotopes are also fissioned by thermal neutrons, σ_{245} 2 020 b and σ_{246} 0.17 b.

^{246}Cm also has a n,γ cross-section of 1.3 b. Because one does not want to loose too much ^{245}Cm, the irradiation is timed to give a maximum yield of ^{246}Cm. If the neutron flux is 2×10^{18} n m^{-2} s^{-1}, (a) when does the ^{246}Cm concentration reach its maximum? (b) What is the ratio between the amount of ^{246}Cm produced and amount of ^{245}Cm consumed at that time?

17.5. A sample containing 50 g ethyl iodide was irradiated with neutrons from a RaBe source of 1×10^9 n m^{-2} s^{-1} for 2 h. If there is a 20% retention and separation occurs 5 min after end of irradiation, what will be the activity of the AgI sample 10 min after separation? The detector has an 8% efficiency for the emitted γ-radiation. Assume the reaction cross-section for ^{127}I(n,γ)^{128}I → (β$^-$, 25.0 min) to be 6.2 b.

17.6. ^{59}Fe of high specific activity can be produced through thermal neutron irradiation of a solution of 1 ml 0.1 M potassium hexacyanoferrate. The recoiling (free) iron atoms, which are produced with a 40% yield, are quantitatively extracted into an organic solvent. How long must the irradiation be in a reactor at 3×10^{19} n m^{-2} s^{-1} to obtain 37 MBq ^{59}Fe? The reaction cross-section for ^{58}Fe(n,γ)^{59}Fe → (β,γ 45.1 d) is 1.15 b and the isotopic abundance of ^{58}Fe is 0.3% in natural iron.

17.9. LITERATURE

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